

Tutorial 1

January 14, 2026

Question 1

Let $C \subseteq \mathbb{R}^n$ be a convex set, with $x_1, \dots, x_k \in C$, and let $\theta_1, \dots, \theta_k \in \mathbb{R}_+$ satisfy $\theta_1 + \dots + \theta_k = 1$. Show that $\theta_1 x_1 + \dots + \theta_k x_k \in C$. (The definition of convexity is that this holds for $k = 2$; you must show it for arbitrary k .) *Hint.* Use induction on k .

Question 2

Assume that for each $\alpha \in A$, C_α is convex, prove that $\bigcap_{\alpha \in A} C_\alpha$ is convex.

Question 3

Show that a set is convex if and only if its intersection with any line is convex.

Question 4

Which of the following sets are convex?

- (a) A *slab*, *i.e.*, a set of the form $\{x \in \mathbb{R}^n \mid \alpha \leq a^T x \leq \beta\}$.
- (b) A *rectangle*, *i.e.*, a set of the form $\{x \in \mathbb{R}^n \mid \alpha_i \leq x_i \leq \beta_i, i = 1, \dots, n\}$. A rectangle is sometimes called a *hyperrectangle* when $n > 2$.
- (c) A *wedge*, *i.e.*, a set of the form $\{x \in \mathbb{R}^n \mid a_1^T x \leq b_1, a_2^T x \leq b_2\}$.
- (d) The set of points closer to a given point than a given set, *i.e.*,

$$\{x \mid \|x - x_0\|_2 \leq \|x - y\|_2, \forall y \in S\}, \text{ where } S \subseteq \mathbb{R}^n.$$

- (e) The set of points closer to one set than another, *i.e.*,

$$\{x \mid \text{dist}(x, S) \leq \text{dist}(x, T)\},$$

where $S, T \subseteq \mathbb{R}^n$, and

$$\text{dist}(x, S) = \inf \{\|x - z\|_2 \mid z \in S\}$$

- (f) The set $\{x \mid x + S_2 \subseteq S_1\}$, where $S_1, S_2 \subseteq \mathbb{R}^n$ with S_1 convex.
- (g) The set of points whose distance to a does not exceed a fixed fraction θ of the distance to b , *i.e.*, the set $\{x \mid \|x - a\|_2 \leq \theta \|x - b\|_2\}$. You can assume $a \neq b$ and $0 \leq \theta \leq 1$.

Question 5

What is the distance between two parallel hyperplanes $\{x \in \mathbb{R}^n \mid a^T x = b_1\}$ and $\{x \in \mathbb{R}^n \mid a^T x = b_2\}$?

Question 6

Which of the following sets S are polyhedra? If possible, express S in the form $S = \{x \mid Ax \preceq b, Fx = g\}$.

- (a) $S = \{y_1 a_1 + y_2 a_2 \mid -1 \leq y_1 \leq 1, -1 \leq y_2 \leq 1\}$, where $a_1, a_2 \in \mathbb{R}^n$.
- (b) $S = \{x \in \mathbb{R}^n \mid x \succeq 0, \mathbf{1}^T x = 1, \sum_{i=1}^n a_i x_i = b_1, \sum_{i=1}^n c_i x_i = b_2\}$, where $a_1, \dots, a_n, c_1, \dots, c_n \in \mathbb{R}$ and $b_1, b_2 \in \mathbb{R}$.
- (c) $S = \{x \in \mathbb{R}^n \mid x \succeq 0, x^T y \leq 1 \text{ for all } y \text{ with } \|y\|_2 = 1\}$.
- (d) $S = \{x \in \mathbb{R}^n \mid x \succeq 0, x^T y \leq 1 \text{ for all } y \text{ with } \sum_{i=1}^n |y_i| = 1\}$.

Question 7

Solution set of a quadratic inequality. Let $C \subseteq \mathbb{R}^n$ be the solution set of a quadratic inequality,

$$C = \{x \in \mathbb{R}^n \mid x^T A x + b^T x + c \leq 0\},$$

with $A \in \mathbb{S}^n$, $b \in \mathbb{R}^n$, and $c \in \mathbb{R}$.

- (a) Show that C is convex if $A \succeq 0$.
- (b) Show that the intersection of C and the hyperplane defined by $g^T x + h = 0$ (where $g \neq 0$) is convex if $A + \lambda g g^T \succeq 0$ for some $\lambda \in \mathbb{R}$.

Are the converses of these statements true?